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CELLULAR FLUIDS

----- Active space -

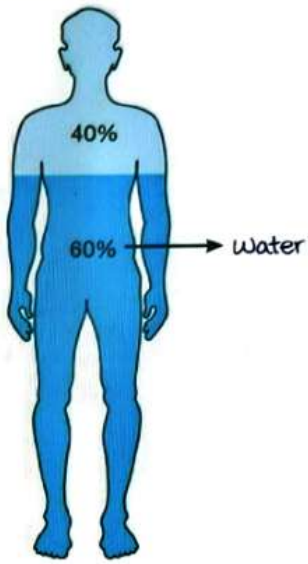
Total Body Water

00:01:00

Maximum contributor of body/cell = Water.

DISTRIBUTION OF BODY WATER :

Total body weight →
- 50-60 % Total body water (TBW).
- 18% Proteins.
- 15% Fats.
- 7% Minerals.



Distribution of body water

Total body water

Intracellular fluid (ICF)
- 2/3rd of TBW.

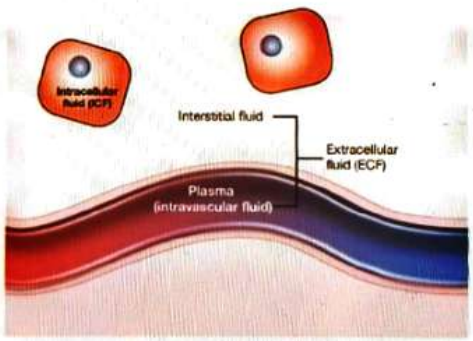
Extracellular fluid (ECF)
- 1/3rd of TBW.

Trans cellular fluid (TF)
- For lubrication.

Plasma
- In blood vessels.

Interstitial fluid (IF)
- In b/w cells.

	Volume	% of body water
TBW	42L	60%
ICF	28L	40%
ECF	14L	20%
Plasma	3.5L	-
IF	10.5L	-
TF	1-2L	-



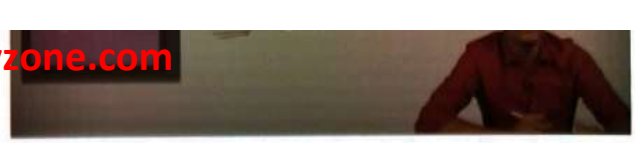
Total body water

Note :
- 75% of total body weight is water in children (males : 60%, females : 50%).
- In diarrhoea → They lose more water → more prone to dehydration.

FACTORS AFFECTING TBW :

	Males	Females
Adipose tissue	↓	↑
Lean body mass	↑	↓
TBW	↑	↓

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Lean body mass (LBM):

- LBM is a form of fat free mass.
- Lean body mass = Body mass – (Adipose tissue + non adipose tissue + bone mass).
- LBM ∝ TBW.
- Water content of LBW = 70 mL/100 g.

Measurements of Body Fluids

00:10:28

Method : Stewart Hamilton method/indicator dilution method.

Procedure :

Inject an indicator into the fluid compartment

↓

measure its concentration in the fluid

Formula :

Amount of fluid injected (A) = Concentration (C) × Volume of the compartment(V)

$$A = C \times V$$

$$V = \frac{A}{C}$$

Modified formula :

$$\text{Volume} = \frac{\text{Amount injected} - \text{Amount excreted}}{\text{Concentration}}$$

INDICATORS & FORMULAS :

Volume	Indicators
Total body water	Deuterium oxide, Tritium oxide
Extracellular fluid	Inulin (m/c), Sucrose, Mannitol, Sodium thiosulphate
Plasma volume (PV)	Albumin (Radiolabelled), Evans blue dye
Blood volume (BV)	Chromium tagged RBCs

Volume	Formulas
Interstitial fluid	No indicators. Calculated by indirect method ECF = PV + IF IF = ECF – PV
Intracellular fluid	No indicators TBW = ECF + ICF ICF = TBW – ECF

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Note :

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1. Formula to calculate blood volume :

$$BV = \frac{PV}{1 - \text{Hematocrit}}$$

2. Blood volume in child = 70 ml/kg.

Fluid Compartments

00:19:00

Fluid compartments contain water and ions.

	ECF	ICF
Dominant cations	Na^+	$K+ > Mg^{2+}$
Dominant anions	Cl^-	Phosphates > Protein anions
pH	7.4	7

WATER LOSS :

Types

- Sensible water loss
  - Sweating (100 ml/day) m/c exercise
- Insensible water loss (600 – 700 ml/day)
  - By evaporation
    - Skin
    - Lung

PLASMA OSMOLARITY :

Plasma osmolarity = 300 mosm/L.

mole : Gram molecular weight of a substance.

Eg : 1 mole of NaCl = 23 + 35.5

= 58.5

Equivalent : $\frac{\text{Gram molecular weight}}{\text{Valency}}$

Eg : $Ca^{2+} = \frac{40}{2} = 20$

Contributors :

1. Na^+ = 270 mosm/L.

major contributor.

2. Proteins = 0.75 mosm/L

Minor contributor due to :

- High molecular weight.
- Low concentration.

TONICITY :

RBC placed in

- Isotonic solution (0.9 % NaCl)
  - No change in shape
- Hypotonic solution (300 mmol/L of urea)
  - Water enters RBC
  - Swells & bursts
- Hypertonic solution (300 mmol/L of mannitol)
  - Water moves out
  - Shrinks

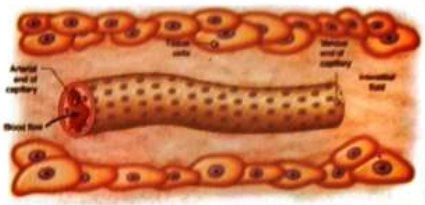
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FACTORS PREVENTING EDEMA :

Edema is accumulation of fluid in interstitial space.



Interstitial fluid compartment

Safety Factor (SF) :

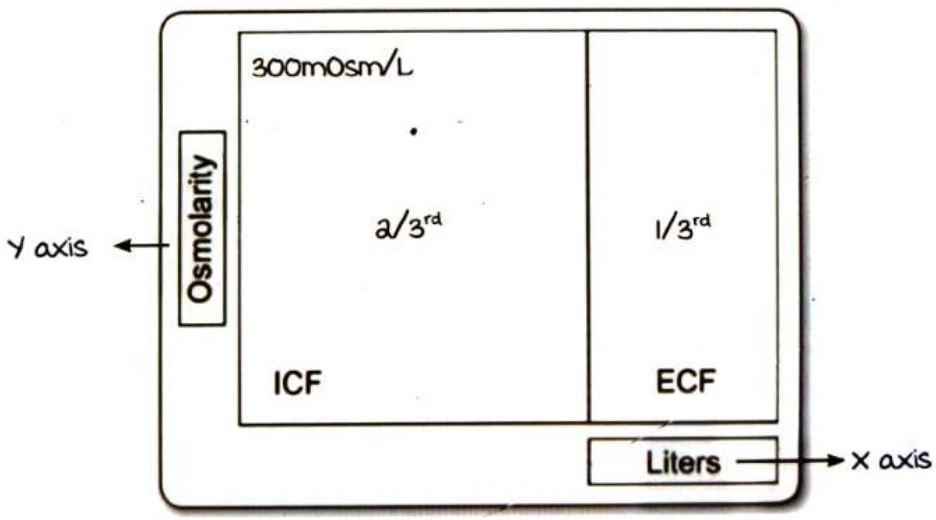
1. Distensibility of the compartment.
   - Normally interstitial space is less distensible.
   - SF = 3 mm of Hg.
2. ↑ Lymphatic flow :
   - Normally, proteins in the blood vessels → Colloid Oncotic pressure.
   - Inflammation → Leakage of proteins from blood vessels.
     Compensatory : ↑ lymphatic flow → moves proteins back inside → Washout of proteins.
     a. ↑ lymphatic flow → SF = 7 mm Hg.
     b. Washout of proteins → SF = 7 mm Hg.
   - Total SF = 3 + 7 + 7 = 17 mm Hg.

Darrow Yannet Diagram

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Alterations in plasma osmolarity & volume → Diseases.

These diseases can be understood via Darrow Yannet diagram.



Normal Darrow Yannet diagram

Saline infusion :

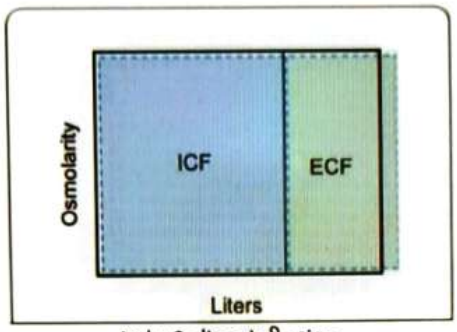
- 0.9% NaCl infusion.
- Treatment of shock.

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	Volume	Osmolarity
ECF	↑	No change
ICF	No change	No change

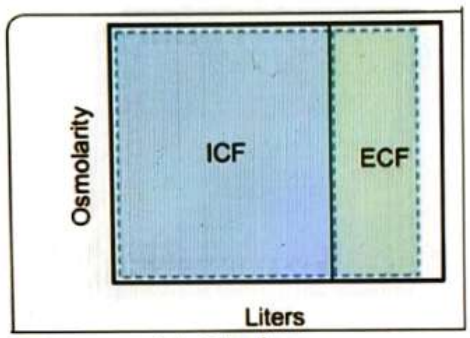


DY : Saline infusion

Diarrhea :

Loss of 0.9% NaCl.

	Volume	Osmolarity
ECF	↓	No change
ICF	No change	No change



DY : Diarrhoea

Hypertension :

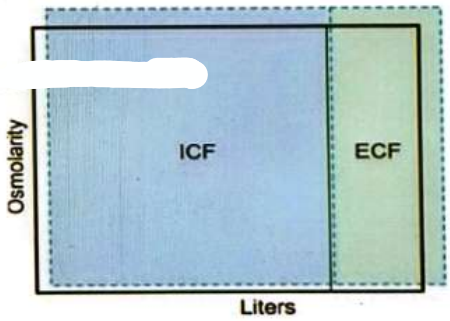
Pathogenesis :

↑ NaCl in ECF (↑ osmolality/hypertonic)

↓

Fluid movement from ICF to ECF

	Volume	Osmolarity
ECF	↑	↑
ICF	↓	↑



DY : Hypertension

Adrenal insufficiency :

Hypoosmotic volume contraction.

Pathogenesis :

Loss of aldosterone

↓

Loss of Na^+

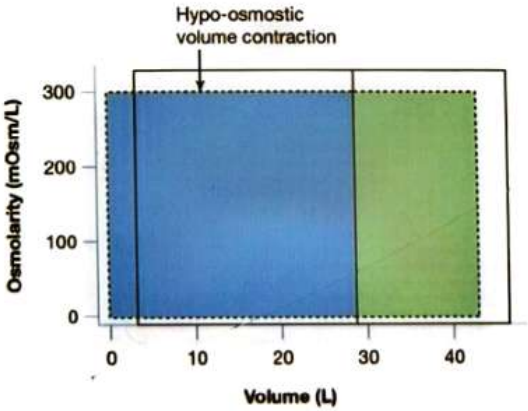
↓

↓ ECF osmolarity

↓

Fluid moves from ECF to ICF

	Volume	Osmolarity
ECF	↓	↓
ICF	↑	↓



DY : Adrenal insufficiency

Feedback